

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
Aptiora–AI Powered Aptitude Learning Platform
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Aptiora–AI Powered Aptitude Learning Platform**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Aptiora is an aptitude learning and practice AI-powered aptitude website, helping all students and job aspirants to prepare for engineering placements, government exams, competitive exams and job interviews. Aptitude, logical reasoning and verbal ability are important skills in many academic or professional fields but there is not such platform to provide all these skills in an organized and interactive order. The Aptiora has been designed to resolve this issue by providing a single platform for a variety of aptitude, logical reasoning, verbal ability and gamified practice questions.

The platform emphasizes the development of problem-solving skills, analysis, quick and accurate responses through continuous practice and interactive learning techniques. It offers a user-friendly and responsive interface, enabling users to effortlessly navigate to various categories of questions and monitor their preparation progress. Gamified learning is incorporated into the learning process which makes it more engaging and motivating for students.

Aptiora is powered by state-of-the-art web technologies and AI-powered features, creating a more enriching learning experience and making aptitude preparation more user-friendly. The project showcases the role of technology and intelligent systems for digital education and skill development. In conclusion, Aptiora is a smart and efficient solution designed to help students and learners enhance their aptitude skills and effectively navigate through different competitive opportunities.

1. Introduction

Aptiora is an aptitude learning and practice platform that uses AI to assist students and aspirants in enhancing their problem-solving and analytical abilities. Aptitude, Logical Reasoning, Verbal Ability are very important in Engineering Placement Test, Government Examinations, Competitive Test, and all kind of Job Interviews in a competitive world. But there are few platforms which offer all these sections in a meaningful and interesting format.

Aptiora is designed to address this issue by providing a single place to find a wide range of aptitude, logical reasoning, verbal ability and gamified practice questions. The primary purpose of Aptiora is to make aptitude preparation easier, smarter and more interactive for the students. The platform offers an intuitive interface that allows users to practice various kinds of questions, increase their speed, accuracy, and boost their general aptitude abilities. The project incorporates gamified learning methods, which make preparation more fun and stress free for the learners. The creation of Aptiora is based on modern technology of web developing and AI components, which allow for creating the individual and efficient learning experiences.

The platform enables the users to track their progress, practice various sets of questions and prepare well for engineering placements, government tests, competitive examinations, and other career opportunities that require aptitude skills. The project emphasizes on the role of digital learning and smart education platforms in the contemporary education system. Aptiora's vision is to connect the smart learning solutions with the traditional approach to preparation for all the aptitude tests and create a place where the students can find the aptitude questions in an organized, easy to use and interactive way.

2. Methodology

The project methodology goes through the following stages:

Aptiora was developed in a structured and user centric approach to build a successful aptitude learning and practice site. The methodology emphasized identification of problem areas in problem solving of the students in the preparation of aptitude and to introduce smart solution which includes learning, practice and engagement in a single platform.

1. Requirement Analysis: The first step was to conduct analysis of the challenges being faced by students in the preparation of aptitude tests, engineering placement tests, government exams and interviews. It was found that the majority of platforms have limited questions or lack organization and interactive learning functions. After analyzing this, the concept of a single aptitude preparation site was born.

2. Platform Design: The interface was designed to be user-friendly and responsive, making it easy to navigate and accessible for users. Various sections like aptitude, logical reasoning, verbal ability and gamified quizzes were set up appropriately to enable the users to practice and develop their skills.

3. Development Process: Modern web technologies were used to build the platform and ensure efficient performance. The system has been designed to include features that enable users to explore a variety of categories of questions, practice frequently and enhance their problem-solving skills. The use of AI-driven ideas and smart features was incorporated to improve the learning experience.

4. Gamification and Interactive Learning: Some gamified features like quizzes, challenges, and interactive learning techniques were added to create a more engaging and motivating learning experience. This will keep the users interested in going through the practice continuously and avoid the stress level typically associated with aptitude preparation.

5. Testing and Improvement: The platform was tested to verify the correct functioning, responsiveness, and smooth user experience. The system was modified in order to enhance the learning outcomes through feedback and performance analysis.

6. Deployment: The platform was successfully tested and deployed as a web-based application, allowing users to easily access it from various devices and prepare anytime, anywhere. In general, the approach of Aptiora was to leverage technology, AI, and interactive learning methods to develop a smart, organized, and effective aptitude preparation platform for students and job aspirants.

3. Tools And Technologies

- **Frontend:** The user interface was developed using Frontend: React.js, TypeScript, Vite, Tailwind CSS, React Router, Radix UI, Recharts, Framer Motion, and Lucide React.
- **Backend:** NodeJS and ExpressJS are used to develop the backend. Prisma ORM is used to manage the SQLite database, bcrypt is utilized for password hashing, Passport.js and Google OAuth 2.0 are utilized for authentication, and Multer is utilized to manage file uploads, while Nodemailer is utilized to send out password reset e-mails.
- **AI / Chatbot:** The Math Mentor chatbot is based on Groq Cloud API and primarily uses LLaMA 3.3 70B as the AI engine. Google Gemini 2.5 Flash is used as the secondary AI engine with a RAG (Retrieval Augmented Generation) pipeline. MongoDB and Mongoose are used to store the Vedic tricks database for RAG.
- **Development & Deployment Tools:** Visual Studio Code was used as code editor.
- **Version Control:** Git and GitHub. The API testing was done using Postman. The frontend is deployed on Vercel, and Prisma Studio was used as a tool to view the database while developing the project.

4. Implementation

Aptiora's implementation was done with a modern full stack web development methodology that leverages technology and artificial intelligence. The objective of the platform was to provide a seamless, interactive, and responsive aptitude learning experience to the learners and job aspirants.

4.1 Frontend Implementation

The front end of Aptiora has been developed with the current JavaScript frameworks and UI libraries, resulting in a beautiful and user-friendly interface.

- React.js components that can be reused and are Dynamic User Interface (UI).
- TypeScript enhanced code reliability and maintainability with the help of static typing.
- Optimized performance and faster development was achieved by using Vite as the frontend build tool.
- Tailwind CSS was used for the responsive and modern UI styling.
- In this app, I have implemented React Router to navigate and route through the pages of every module of the platform.
- Some components were also added for more accessibility and usability, such as Radix UI components.
- Framer Motion was used to create smooth transitions and animations throughout the app.
- Recharts assisted with showing the progress and analytics of users graphically.

The user-friendly frontend display was built to make sure that users can enjoy a smooth experience across all devices, whether they are using a desktop, tablet or phone.

4.2 Backend Implementation

The back-end of Aptiora is built with Node.js and Express.js, ensuring efficient API management, authentication, and application logic.

- Express.js was used to create RESTful APIs for the front and back end to communicate.
- With the help of Prisma ORM, the management and handling of databases and queries with SQLite were made easier.

- The reason for implementation of bcrypt in the system was to ensure secure password hashing and protect the users' data.
- Secure authentication and social login were achieved by integrating Passport.js and Google OAuth 2.0.
- Multer supported features to upload files and process documents.
- For sending password reset and verification e-mails to users, Nodemailer was used.

The backend architecture was designed to handle the queries efficiently, secure data management and smooth communication between various modules of the system.

4.3: Integrate AI and Chatbots Workplace

One of the key highlights of Aptiora is its Math Mentor Chatbot, which offers AI-powered learning assistance.

- The chatbot is powered by the Groq Cloud API and the LLaMA 3.3 70B AI model is the main model used.
- Google Gemini 2.5 Flash is now available as a second AI engine to use for improved response generation and fallbacks.
- A RAG (Retrieval Augmented Generation) pipeline was driven for giving accurate and context-aware explanations of the concepts of aptitude and the Vedic Mathematics tricks.
- During the interaction with the chatbot, it was important that these Vedic tricks be retrieved efficiently, hence MongoDB and Mongoose were used to store and manage the database of Vedic tricks.

The AI module guides the user to understand the problem solving approach, boost the speed of calculation and also provides immediate feedback while solving aptitude problems.

4.4 Database Implementation

The platform is equipped with several databases to ensure optimal data management.

- All data about the users, quizzes, progress tracking, authentication, platform etc. is kept in an SQLite database using Prisma ORM.

- The RAG-based chatbot system stores the AI related learning resources and content in MongoDB.
- Database design was tailored to data retrieval, scalability and user and learning data management.

4.5 Gamification Features

To maintain the learning of aptitude interesting and not boring in nature, the following features were added to the Aptitora, which are gamified features. The following gamified features are added to aptitude learning, so that the learning of aptitude is not boring and less stressful. To make the learning of Aptitude more interesting and less stressful Aptitora was enhanced with the following gamifying features:

- Interactive quizzes and timed challenges to test knowledge and skills.
- Periodic performance score tracking and assessment.
- Progress monitoring (in charts and graphs)

The features increase the interest of the user and encourage regular use.

4.6 Testing and Deployment

The platform has been well-tested for features and functions, as well as for responsiveness, security, and functionality.

- The API testing and backend validation was performed using Postman.
- The device responsiveness testing process to cross devices ensures across compatibility.
- During the testing process, bug fixes and performance improvements were implemented.

For deployment:

The frontend application is hosted using Vercel with an aim to provide fast and reliable hosting.

The backend is deployed using Render and database with Supabase.

A version control system and collaborative development was conducted with git and GitHub.

5. Testing and Validation

Testing and validation were a crucial part of the development of Aptiora, as it is essential to ensure that the platform operates smoothly, securely, and efficiently on a variety of devices and scenarios. Different tests were conducted to ensure performance, reactivity and reliability of the system.

5.1 Functional Testing

Functional testing was performed to check the functioning of all modules and features of the platform as per the requirement. These are the functionalities that were tested:

- Sign up and log in.
- Google OAuth Authentication
- Three modules for Aptitude, Logical Reasoning and Verbal Ability. Three question modules: Aptitude, Logical Reasoning and Verbal Ability.
- The AI Chatbot Gives a Response and Interacts with the User
- Timed Challenges and Gamified Quiz Features.
- Progress tracking and performance analysis.
- Resetting passwords and verifying emails. System for resetting passwords and verifying email.

All the essential functions were tested and outcomes were obtained as expected.

5.2 API Testing

API testing was conducted by using Postman to ensure that the communication between the Frontend and Backend systems is correct. Different API endpoints were tested for:

- The process of fetching and submitting data.
- Authentication and authorization
- Error handling and validation.
- Secure response handling

APIs responded appropriately and data was handled correctly, including proper status codes.

5.3 User Interface Testing

The user interface is tested and proved and is user-friendly. Testing focused on:

- Design that adapts to the desktop, tablet and cell phone screen size. Design that adjusts to the desktop, tablet and cell phone screen size.
- The ability to move from one page to another and/or one section to another.
- The consistency and accessibility of the UI. Consistency and accessibility of the UI.
- Correct animation, charts and interactive features

The interface was responsive on different screen sizes and browsers and looked the same.

5.4 Database Testing

Efficient Data storage, retrieval and management were verified with database testing.

- The users' information and the quiz results were successfully validated.
- The query accuracy and query speed were checked in databases.
- In the RAG pipeline, the data and Vedic tricks related to AI were verified to be retrieved correctly.
- Prisma ORM operations validated for smooth relationship with database
- The database system remained in good integrity and performance during tests.

5.5 Security Testing

Security testing was carried out to safeguard user information and to ensure secure access to the platform.

- All passwords were securely encrypted using the bcrypt method. All passwords were encrypted securely by bcrypt.
- Passport.js and Google OAuth authentication were tested.
- Unauthorized access prevention, session handling was tested.
- Input validation was implemented to reduce security vulnerabilities

The platform proved secure authentication and safe management of users' information.

5.6 Performance Testing

Performance testing was done to check the speed and stability of the platform at normal usage.

- Frontend optimization, using Vite and modern practices, for page loading speed.

- The API response times were tracked and optimized on the back end.
- Cloud deployment performance was tested on Vercel and Render.
- The performance was benchmarked on Supabase PostgreSQL database.

The system was stable, user interactions smooth and there was good response time.

5.7 Validation Results

Following the testing processes, Aptitora was found to be a functional and reliable aptitude learning platform. The system met its goals of:

- Organized aptitude preparation.
- Providing AI-based learning support services
- Providing motivating, gamified practice opportunities
- Ensuring the performance of a system was secure and responsive.

The testing and validation stage proved that Aptitora is capable of offering an effective, easy to use and scalable aptitude learning and competitive exam preparation solution.

6. Results & Discussion

As a result of aptiora's development and implementation, an AI powered aptitude learning and practice platform was successfully created for learners and job aspirants. The platform brings together all the components of aptitude preparation, interactive learning, gamification, and AI support under a unified and organized umbrella.

6.1 Results Obtained

The project goals were achieved and the following outcomes were achieved with the final system:

- A very good aptitude preparation web platform was developed without any problem that is responsive and user friendly.
- Users will be able to take out the questions of Aptitude, Logical Reasoning and Verbal Ability in a structured way easily.
- Users were able to better grasp concepts and problem-solving skills through the use of AI-powered chatbot support.
- The incorporation of gamified quizzes and interactive challenges enhanced engagement and motivation among users.
- Security of authentication and account management was implemented successfully.
- It contained progress monitoring and performance analysis, allowing users to monitor their performance throughout time.
- The platform has been released in the internet successfully, by utilizing modern cloud hosting technologies.

The platform's functionality and learning experience were enhanced by utilizing the integration of modern web technologies and AI tools.

6.2 Discussion

The project showcases the advantages of using modern technology and intelligent systems in the context of enhancing digital education and aptitude training. The traditional methods of learning aptitude are more repetitive and less interactive leading to lesser involvement by the students. Aptiora tackles these problems by implementing AI-powered support and gamification strategies.

This Math Mentor chatbot, with artificial intelligence, provided immediate assistance and concept explanations, enhancing the maths learning experience with a more interactive and personalized

approach. Responses were also improved in terms of accuracy and context understanding through the use of Retrieval Augmented Generation (RAG).

The front-end was also designed to be responsive, using the React.js library and Tailwind CSS, making the site more accessible and usable on different devices, and allowing users to practice whenever and wherever they wanted. Seamless integration of data handling and secure interactions between the system's components was achieved by leveraging Node.js, Express.js, Prisma ORM, and Supabase.

Aptitude preparation has also been lightened up by the use of gamification elements, which have helped users 'engage' and get the stress off. Ongoing assessment and learning through quizzes, challenges and progress monitoring.

During the development process, some challenges were faced in the integration of AI models, response handling from the API, and smooth communication between the front-end and back-end services. The problems were solved by testing, optimizing and applying correct API management strategies.

To conclude, Aptitora has accomplished its mission objective in preparing a smart, interactive and efficient aptitude preparation platform. The project underscores the pivotal role of AI in enhancing educational experiences and aiding students in competitive settings.

7. Conclusion And Future Scope

Aptitora is a smart and interactive aptitude learning platform which have been designed to study and prepare students and job aspirants well for engineering placement tests, government exams, competitive exams and interviews. The project has been able to merge the concepts of aptitude, logical reasoning, verbal ability with gamified learning on a single platform making the preparation more organized, accessible and enjoyable. With its up-to-date web technology and AI features, Aptiora enhances users' problem-solving skills, time efficiency, accuracy, and confidence by providing them with ongoing practice and interactive learning opportunities. The platform demonstrates the importance of digital education and intelligent learning systems in today's competitive environment.

It offers a user-friendly environment to which the students can easily go to a large number of questions and enhance their aptitude skills in an efficient way. In the overall perspective, Aptiora is an impressive platform for providing students with comprehensive aptitude preparation. The platform can be expanded in the future with sophisticated AI-powered recommendations, individual learning routes, mock exams, instant assessment and analysis of performance and company-specific modules for aptitude preparation. Other features like leader boards, achievements, real-time quizzes, discussion forums and support in multiple languages can help enhance user engagement and accessibility.

An increase of gamification techniques with greater interaction and mobile application support can also be added to make learning more flexible and fun. As Aptiora evolves and embraces innovation, it could become a comprehensive and intelligent aptitude preparation platform for students in various educational and career domains.

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